

SafetyEye — PPE Violation System

AI-Based Helmet, Mask & Vest Violation Detection Using YOLOv8

1. Abstract

SafetyEye is an advanced AI-powered monitoring system designed to automate Personal Protective Equipment (PPE) compliance in workplaces. Using the YOLOv8 deep learning model, the system can detect safety violations—such as the absence of helmets, masks, or safety vests—in real time from live video streams.

The system automatically logs violation details, captures visual evidence, and generates a PDF challan report for record-keeping. It also sends instant email alerts with attachments to supervisors. A Streamlit-based dashboard provides real-time monitoring, analytics, reporting, and system configuration options.

This project significantly enhances workplace safety management by reducing manual monitoring efforts and improving violation traceability.

2. Problem Statement

Industries like construction, manufacturing, and mining rely heavily on strict PPE compliance to ensure worker safety. Traditional manual monitoring of PPE usage has several drawbacks:

- ⚠️ **Human error and inconsistent supervision**
- ⚠️ **High operational cost** for deploying multiple supervisors
- ⚠️ **Difficulty monitoring large or multi-camera workplaces**
- ⚠️ **Delayed reporting and insufficient documentation**

There is a critical need for an **automated, real-time, scalable, and accurate system** to identify PPE violations and maintain proper audit-ready evidence.

3. Objectives

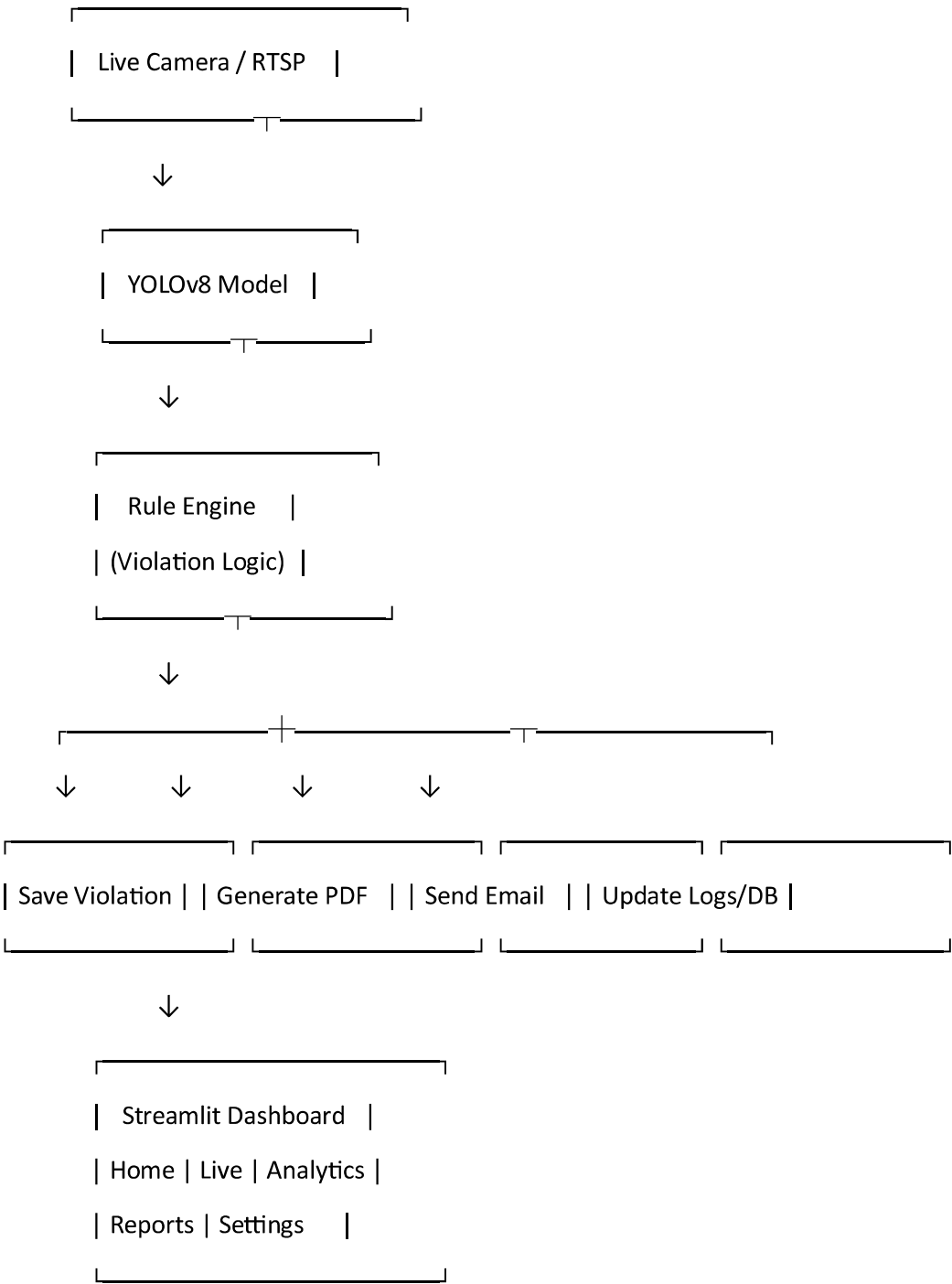
The primary goal of SafetyEye is to automate PPE violation monitoring.

The system aims to:

- ✓ Detect PPE violations (NO-Hardhat, NO-Mask, NO-Vest) in real time
- ✓ Capture and store image evidence of each violation
- ✓ Generate downloadable PDF challans with details
- ✓ Send automated email alerts with attachments

- ✓ Provide an interactive dashboard for visualization
 - ✓ Track historical data and trends
 - ✓ Allow administrators to adjust detection settings and alert configurations
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4. System Architecture



5. Dataset Description

The model is trained on the **Construction Site Safety Image Dataset** available on Roboflow/Kaggle. It contains images of workers with and without PPE items.

Classes Used

- Hardhat
- Mask
- Vest
- NO-Hardhat
- NO-Mask
- NO-Vest

Dataset Split

- 70% — Training
- 20% — Validation
- 10% — Testing

All annotations are in **YOLO format**, ensuring smooth integration.

6. Model Training – YOLOv8

YOLOv8 is chosen due to:

- ⚡ High real-time inference speed
- 🎯 Excellent accuracy
- ☐ Easy integration with Python
- ✂️ Pretrained model availability

Training Workflow

1. Prepare dataset in YOLO format
2. Configure data.yaml
3. Train the model on Kaggle/Colab
4. Evaluate performance on test data
5. Export and use yolov8s.pt in the system

Sample Training Command

```
yolo train model=yolov8s.pt data=data.yaml epochs=50 imgsz=640
```

Final Model Used

7. Backend Components

a) detector.py

- Loads YOLO model
- Performs frame-by-frame detection
- Extracts class names & confidence
- Saves violation images
- Writes logs to violations_log.csv

b) utils.py

- Generates PDF challan using ReportLab
- Handles file operations
- Supports image rendering in PDFs

c) email_alert.py

- Sends real-time email alerts
- Supports attachments (image/PDF)
- Uses Gmail SMTP + App Password

d) live_monitor.py

Manages the real-time monitoring pipeline:

- Captures webcam/RTSP stream
- Runs YOLO inference
- Applies rule engine
- Generates alerts
- Triggers email + PDF
- Displays frame with annotations

8. Frontend (Streamlit Dashboard)

◆ a) Home

- Displays total violations
- Shows recent records
- Quick overview of system status

◆ b) Live Monitor

- Streams webcam in real time
- Highlights PPE/non-PPE detections
- Shows FPS, violation count
- Generates PDF challans

◆ c) Analytics

Interactive visualizations:

- KPI cards
- Pie chart for violation distribution
- Trend line over time
- Most common violations

◆ d) Reports

- Displays all recorded violations
- Allows CSV export
- View/download generated PDFs

◆ e) Settings

- Configure detection threshold
- Enable/disable audio alerts
- Email sender/receiver/app-password
- Toggle PDF generation
- Saves to settings.json

9. Database

Stored in /database/:

- violations_log.csv — master log
- exported_violations.csv — downloadable reports
- settings.json — system configuration
- challans/ — PDFs generated with evidence

Log Format

id, label, confidence, image_path, timestamp

10. Results

The SafetyEye system successfully:

- Detects PPE and non-PPE conditions accurately
 - Identifies Helmet/Mask/Vest violations
 - Generates high-quality PDF challans
 - Sends automatic email alerts
 - Logs all violations with traceable IDs
 - Provides an analytics dashboard for insights
 - Maintains complete report history
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11. Conclusion

SafetyEye demonstrates how AI and computer vision can significantly improve workplace safety compliance. By automating PPE monitoring, it reduces manual supervision effort, increases accuracy, and ensures proper documentation through PDFs and logs.

With real-time monitoring, alerting, and detailed analytics, SafetyEye is a practical and scalable solution for industries requiring strict safety protocols.

12. Future Scope

To further enhance the system, the following features can be added:

- 🔍 Worker identification using facial recognition
- 📱 Mobile app for supervisors
- ☁️ Cloud database with remote monitoring
- 📷 Multi-camera and 360° view support
- 💬 WhatsApp or SMS alert system
- ⌚ Behavior tracking / worker attendance
- 📊 Advanced analytics using machine learning